

Program 7 : Deploy a Web Application to Kubernetes using Minikube

STEP 1 - Start Minikube

minikube start

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$ ls
nginx.deployment.yml
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
🚚 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, dashboard, default-storageclass

! /usr/bin/kubectl is version 1.28.15, which may have incompatibilities with Kubernetes 1.34.0.
  ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2 : Check status

minikube status

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

STEP 3 - If Minikube is not running, restart it

minikube delete

minikube start --driver=docker

STEP 4 - Create the Deployment file (nginx-deployment.yaml)

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$ cat nginx.deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$
```

STEP 5 - Apply the deployment

kubectl apply -f nginx-deployment.yaml

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$ k apply -f nginx.deployment.yml
deployment.apps/nginx-deployment created
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$
```

STEP 6 - Check Deployment & Pods

kubectl get deployments

kubectl get pods

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$ kubectl get deployments
kubectl get pods
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  2/2     2            2           60s
NAME          READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-c78kt  1/1     Running   0          60s
nginx-deployment-6f9664446b-va5k8  1/1     Running   0          60s
```

STEP 7-Expose the Deployment as a Service

kubectl expose deployment nginx-deployment --type=NodePort --port=80

This creates a service that exposes Nginx to your system.

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$ k expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
```

STEP 8-Check Services

`kubectl get services`

A service named `nginx-deployment` should appear.

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$ kubectl get services
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	21d
nginx-deployment	NodePort	10.100.89.220	<none>	80:30872/TCP	46s

STEP 9-Access the Webpage

Minikube provides a shortcut to open the app:

`minikube service nginx-deployment`.

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30872

🌐 Opening service `default/nginx-deployment` in default browser...

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab7$ Opening in existing browser session.
```

